codes.bsn

Purpose

`codes.bsn` holds the method-selection switches that apply to the whole basin: the PET method, the routing method, the carbon model, the soil P model, and so on. It is read into `bsn_cc` (`type basin_control_codes`) early in the run and then consulted by every process module.

Source

- Reader: src/basin_read_cc.f90
- Type definition: src/basin_module.f90, `type basin_control_codes`

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: 26 fields, list-directed, read into `bsn_cc` in declaration order.

#	Field	Type	Default	Description
1	petfile	char(16)	`pet.cli`	Potential ET filename. Used when `pet = 3` (read PET from file).
2	wwqfile	char(16)	`""`	Watershed stream water-quality filename.
3	pet	int	0	PET method. `0` Priestley-Taylor, `1` Penman-Monteith, `2` Hargreaves. (Method `3` triggers reading `pet.cli`.)
4	nam1	int	0	Not used.
5	crk	int	0	Crack flow. `1` to compute flow in cracks.
6	swift_out	int	0	SWIFT output. `0` do not write, `1` write `swift_hru.inp`.
7	sed_det	int	0	Peak rate method. `0` NRCS dimensionless hydrograph with PRF, `1` half-hour rainfall intensity method.
8	rte	int	0	Water routing method. `0` variable storage, `1` Muskingum.
9	deg	int	0	Not used.
10	wq	int	0	Not used.
11	nostress	int	0	Plant stress. `0` all stresses applied, `1` turn off all plant stress, `2` turn off nutrient stress only.
12	cn	int	0	Not used.
13	cfac	int	0	Not used.
14	cswat	int	0	Carbon model. `0` static (legacy mineralization), `1` C-FARM (reserved, not implemented), `2` dynamic CENTURY/SWAT-C model.
15	lapse	int	0	Lapse-rate adjustment for precip and temperature. `0` no, `1` yes.
16	uhyd	int	1	Unit hydrograph. `0` triangular, `1` gamma function.
17	sed_ch	int	0	Not used.
18	tdrn	int	0	Tile drainage equation. `0` drawdown-days, `1` DRAINMOD.

#	Field	Type	Default	Description
19	wtdn	int	0	Shallow water-table depth. `0` original routine (fill to upper limit), `1` DRAINMOD water-table depth.
20	sol_p_model	int	0	Soil P model. `0` original SWAT P model, `1` Vadas and White (2010).
21	gampt	int	0	Infiltration method. `0` curve number, `1` Green-Ampt.
22	atmo	char(1)	`a`	Not used.
23	smax	int	0	Not used.
24	qual2e	int	0	Instream nutrient routing. `0` QUAL2E, `1` QUAL2E with simplified transformations.
25	gwflow	int	0	gwflow module. `0` inactive, `1` active. Triggers extra logic in `object.cnt`.
26	idc_till	int	3	Tillage method when `cswat = 2`. `1` DSSAT, `2` EPIC, `3` Kemanian, `4` DNDC.

The "Not used" fields are read from the file but the model does not consult them. Tools commonly write `0`.

Side effects after read

If `pet = 3`, the reader opens `pet.cli` and reads a title and header line to position the file pointer. The actual PET values are read later by the climate routines.

Example

`refdata/Ames_sub1/codes.bsn`:

`codes.bsn`:

pet_file	wq_file	pet	event	crack	swift_out	sed_det	rte_cha	deg_cha
null	null	2	0	0	0	0	0	0

The header column names written by SWAT+ Editor (`event`, `rte_cha`, `i_fpwet`, …) differ from the field names in the source (`crk`, `rte`, `qual2e`, …). Order is what the reader cares about. Mapping the example to the table:

- `pet = 2`: Hargreaves.
- `cswat = 2`: dynamic CENTURY/SWAT-C carbon model. (Header label `carbon`.) The example file above still carries `carbon = 1`, which is now the reserved C-FARM slot and runs no carbon model; datasets that want the dynamic model must set `carbon = 2`.
- `uhyd = 1`: gamma-function unit hydrograph.
- `idc_till = 3`: Kemanian tillage method.

The `null` strings in columns 1 and 2 disable the PET and water-quality file inputs.

Related files

- `parameters.bsn`. Numeric basin parameters consumed by the methods selected here.
- `object.cnt`. Reads gwflow state when `gwflow = 1`.
- `pet.cli`. Used when `pet = 3`.